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ELECTRICAL CABLE FOR THE CONNECTION OF ELECTRIC/ELECTRONIC DEVICES, IN PARTICULAR ROAD VEHICLES

Abstract

An electrical cable for the connection of two electric/electronic devices, in particular road vehicles, is defined by at least one electrical conductor inserted in a protective sheath made of an insulating material having two sheath portions, which define, together with the segments of electrical conductor associated thereto, relative segments of electrical cable, and are separated from one another so as to enable the two segments of electrical cable to rotate with one another about a longitudinal axis of the electrical cable.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS

[0001] This patent application claims priority from Italian patent application no. 102024000004696 filed on Mar. 4, 2024, the entire disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

[0002] The present invention relates to an electrical cable for the connection of electric/electronic devices, in particular road vehicles.

BACKGROUND

[0003] In the field of road vehicles, it is known to provide an electrical cable to connect a first electric/electronic device, e.g. a battery pack, and a second electric/electronic device, e.g. a charging socket, to one another.

[0004] Generally, the electrical cable comprises at least one electrical conductor and a protection sheath made of insulating material fitted around the electrical conductor.

[0005] Known electrical cables of the type described above have certain disadvantages mainly resulting from the fact that these electrical cables only allow to connect fixed electric/electronic devices to one another and are unable to connect mobile electric/electronic devices to one another such as, for example, a battery pack and a charging drum provided with a plurality of charging sockets and movable so as to allow the selective connection of the charging sockets with a source of electrical energy external to the road vehicle.

SUMMARY

[0006] The object of the present invention is to provide an electrical cable for the connection of electric/electronic devices, in particular road vehicles, which is free from the above-mentioned drawbacks and which is of simple and cost-effective production.

[0007] According to the present invention, there is provided an electrical cable for the connection of electric/electronic devices, in particular road vehicles, as claimed in claims **1** to **6**.

[0008] The present invention further relates to an electrical connection system for electric/electronic devices, in particular road vehicles.

[0009] According to the present invention, there is provided an electrical connection system for electric/electronic devices, in particular road vehicles, as claimed in claims **7** to **10**.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The present invention will now be described with reference to the attached drawings, which show a non-limiting embodiment thereof, wherein:

[0011] FIG. **1** is a schematic perspective view, with parts removed for clarity, of a preferred embodiment of the electrical cable of the present invention implemented in an electrical connection system for electric/electronic devices;

[0012] FIG. **2** is an exploded schematic perspective view, with parts removed for clarity, of the electrical connection system in FIG. **1**;

[0013] FIG. **3** is a schematic perspective view of the electrical cable in FIGS. **1** and **2**;

[0014] FIG. **4** is a schematic perspective view of a first variant of the electrical cable in FIG. **1**; and

[0015] FIG. **5** is a schematic perspective view of a second variant of the electrical cable in FIG. **1**.

DESCRIPTION OF EMBODIMENTS

[0016] With reference to FIGS. **1** and **2**, **1** denotes, as a whole, an electrical connection system comprising two electric/electronic devices **2**, **3** and a connection device **4** to connect devices **2**, **3** to one another.

[0017] For example, the device **3** is a battery pack for powering an electric motor of an electric or hybrid road vehicle, and the device **2** is a charging drum, which is provided with a plurality of charging sockets, and is movable to allow the selective connection of the charging sockets with a source of electrical energy external to the road vehicle.

[0018] The system **1** further comprises a container **5** to accommodate the device **2** therein, and a rolling bearing **6** mounted through a wall **7** of the container **5**.

[0019] Obviously, according to a variant not shown, the device **3** is fixed and the assembly defined by the device **2** and the container **5** is movable relative to the device **3**.

[0020] The connection device **4** comprises an electrical cable **8** comprising, in turn, in this case, a plurality of conductor wires **9**, in this case three wires **9**, and an outer protection sheath **10** made of insulating material fixed around the wires **9**.

[0021] According to a variant not shown, the cable **8** is a shielded cable with a metal element arranged between the wires **9** and the sheath **10**.

[0022] According to a further variant not shown, the wires **9** are covered with a layer of highly elastic insulating material.

[0023] Each wire **9** comprises an electrical conductor **11** and an outer protection sheath **12** made of insulating material fixed around the conductor **11**.

[0024] The sheath **10** comprises two sheath portions **13**, **14**, which define, together with the segments **9** of the wires associated with the portions **13**, **14**, relative segments **15**, **16** of cable **8**, and are separated from one another so as to allow the two segments **15**, **16** of cable **8** to rotate relative to one another around a longitudinal axis **17** of the cable **8**.

[0025] The segment **15** extends inside the container **5** and is connected with the device **2**; the segment **16** extends substantially outside the container **5** and is connected with the device **3**; and the sheath portion **13** is coupled in an angularly fixed manner to an inner track of the bearing **6** through the interposition of a cable gland **18** mounted through the bearing **6**.

[0026] With regard to what set forth above, it should be noted that: [0027] the wires **9** comprise a free portion **19**, which extends between the two portions **13**, **14**, and is substantially accommodated inside the container **5**; and [0028] the portions **13**, **14** are arranged at a distance H from each other, measured parallel to the axis **17**, which is greater than an outer diameter D**1** of the sheath **10** and than an outer diameter D**2** of each sheath **12** (FIG. **3**).

[0029] The variant shown in FIG. **4** differs from what shown in the previous figures only in that, in it, the electrical cable **8** is eliminated and replaced with an electrical cable **20** comprising an outer protection sheath **21** made of insulating material attached around an electrical conductor **22** defined by a single bundle **23** of metal wires extending through the sheath **21** itself.

[0030] The sheath **21** comprises two sheath portions **24**, **25**, which define, together with the segments of conductor **22** that are associated with the portions **24**, **25**, relative segments **26**, **27** of cable **20**, and are separated from one another so as to allow the two segments **26**, **27** of cable **20** to rotate relative to one another around a longitudinal axis **28** of the cable **20** itself.

[0031] The portions **24**, **25** are arranged at a distance H greater than an outer diameter D**3** of the sheath **21** and than a diameter D**4** of the conductor **22**.

[0032] The variant shown in FIG. **5** differs from what shown in FIG. **4** only in that, in it, the conductor **22** comprises a plurality of metal wires grouped in a single bundle **29** at each sheath portion **24**, **25** and in at least two bundles **30** (in this case three bundles **30**) between the two sheath portions **24**, **25**.

[0033] The distance H between the portions **24**, **25** is greater than the outer diameter D**3** of the sheath **21**, than a diameter D**5** of the bundle **29**, and than a diameter D**6** of each bundle **30**.

[0034] The electrical connection system **1** has certain advantages mainly resulting from the fact that the electrical cables **8**, **20** allow to connect electric/electronic devices **2**, **3** movable relative to one another.

Claims

1. An electrical cable for the connection of electric/electronic devices, in particular road vehicles, the electrical cable comprising at least one electrical conductor to electrically connect a first electric/electronic device and a second electric/electronic device to one another; and a first protection sheath made of an insulating material and fixed around the electrical conductor; and characterized in that the first protection sheath comprises two sheath portions, which define, together with the segments of electrical conductor associated with the sheath portions, relative segments of electrical cable and are separate from one another so as to allow the two segments of electrical cable to rotate relative to one another around a longitudinal axis of the electrical cable.
2. The electrical cable according to claim 1, wherein the two sheath portions are arranged at a distance from one another that is greater than an outer diameter of the first protection sheath and than a diameter of each electrical conductor.
3. The electrical cable according to claim 1, wherein the electrical conductor comprises on single bundle of metal wires extending through the entire first protection sheath.
4. The electrical cable according to claim 1, wherein the electrical conductor comprises a plurality of metal wires grouped in one single bundle in the area of each sheath portion and in at least two bundles between the two sheath portions.
5. The electrical cable according to claim 1, and comprising at least two conductor wires, each comprising, in turn, an electrical conductor and a second protection sheath (**12**) made of an insulating material and fixed around the electrical conductor; the conductor wires extending through the first protection sheath.
6. The electrical cable according to claim 5, wherein the two sheath portions are arranged at a distance from one another that is greater than an outer diameter of the first protection sheath and than a diameter of each conductor wire.
7. An electrical connection system for electric/electronic devices, in particular road vehicles, the electrical connection system comprising a first electric/electronic device; a second electric/electronic device; and an electrical cable made according to claim 1 for electrically connecting said first and second electric/electronic devices to one other.
8. The electrical connection system according to claim 7, and further comprising a container to accommodate, on the inside, the first electric/electronic device and a rolling bearing fitted through a wall of the container; a first segment of electrical cable being fixed to the first electric/electronic device and a second segment of electrical cable being fixed to the second electric/electronic device and to the rolling bearing so as to rotate relative to the first segment of electrical cable around said longitudinal axis.
9. The electrical connection system according to claim 8, wherein the second segment of electrical cable is fixed to the rolling bearing through the interposition of a cable gland.
10. The electrical connection system according to claim 8, wherein the electrical conductor portion arranged between the two sheath portions extends inside the container.